Detailed Insights Report on Selected Attributes

Part 1

Objective: To explore the relationship between the **class variable Credit_Score** (Good, Standard, Poor) (Good, Standard, Poor) and various **numeric attributes** based on mean differences, statistical significance (p-value), and the potential implications of these differences.

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1. Age

- **Insight:** Customers with a Good credit score have a higher mean age (36.40 years) compared to those with Standard (33.43 years) and Poor (31.19 years) credit scores.
- **Statistical Significance:** The p-value is < 0.0001, indicating a highly significant difference in the mean ages across credit score categories.
- **Implication:** Older customers may have more experience in managing finances, which could lead to better credit scores. This suggests that age is a possible indicator of financial maturity and stability, which in turn may help in achieving a better credit score.

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2. Annual Income

- **Insight:** The average annual income increases with the credit score category. Customers with a Good credit score earn an average of **$65,203.67**, significantly more than those with Standard (**$50,987.16**) or Poor (**$40,584.52**) credit scores.
- **Statistical Significance:** The p-value is < 0.0001, confirming a significant relationship between credit score and annual income.
- **Implication:** Higher income is strongly associated with better credit scores. A higher income likely provides individuals with more disposable income, making it easier for them to meet payment obligations, which positively impacts credit scores.

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3. Changed Credit Limit (%)

- **Insight:** The Good credit score category has the lowest average percentage change in credit limit (**7.10%**) compared to Standard (**11.13%**) and Poor (**11.05%**) credit scores.
- **Statistical Significance:** The p-value is < 0.0001, suggesting that the difference in credit limit changes is statistically significant.
- **Implication:** A higher percentage change in credit limit may suggest that lenders are attempting to adjust credit limits in response to perceived financial instability, which is more common among individuals with lower credit scores. Those with higher scores may not require as many adjustments, implying stability in their financial habits.

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4. Credit History Age (months)

- **Insight:** Customers with a Good credit score have the longest credit history (284.75 months), followed by those with Standard (227.80 months) and Poor (170.05 months) credit scores.
- **Statistical Significance:** The p-value is < 0.0001, indicating a strong and significant relationship between credit history length and credit score.
- **Implication:** A longer credit history provides more data for lenders to assess the customer's creditworthiness. Individuals with a longer credit history are generally perceived as less risky, and thus, are more likely to have a Good credit score. This supports the notion that a longer financial history aids in building a better credit score.

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5. Delay from Due Date (days)

- **Insight:** There is a strong difference in the average number of days delayed from the payment due date. Customers with a Poor credit score average **29.74** days of delay, while those with a Standard score average **19.75** days, and those with a Good score average **10.89** days.
- **Statistical Significance:** The p-value is < 0.0001, indicating that the differences in delay from the due date across credit score categories are highly significant.
- **Implication:** Delays in payment are a key factor in credit score calculation. Customers with a Good credit score show the least delay, suggesting they are more likely to pay bills on time. Those with a Poor credit score experience frequent delays, which negatively impacts their credit scores. Timely payment is critical to maintaining a good credit score.

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6. Interest Rate (% per annum)

- **Insight:** Interest rates for credit card holders are lowest for customers with a Good credit score (7.66%), moderate for those with a Standard score (13.75%), and highest for those with a Poor score (20.19%).
- **Statistical Significance:** The p-value is < 0.0001, highlighting that the relationship between credit score and interest rates is significant.
- **Implication:** A lower credit score generally results in higher interest rates because lenders perceive those individuals as higher risk. This reinforces the idea that managing credit responsibly—reflected by the credit score—can lower borrowing costs.

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7. Monthly In-hand Salary (USD)

- **Insight:** Customers with a Good credit score earn a higher monthly in-hand salary on average (**$5,403.11**) compared to Standard (**$4,240.74**) and Poor (**$3,381.39**) credit score holders.

- **Statistical Significance:** The p-value is < 0.0001, suggesting a significant relationship between monthly salary and credit score.
- **Implication:** A higher income provides individuals with more resources to meet their credit obligations, which is likely to contribute to better credit scores. Financial stability and the ability to manage credit effectively are often linked to higher salaries.

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8. Number of Bank Accounts

- **Insight:** Customers with a Good credit score tend to have fewer bank accounts (**3.44**) compared to Standard (**5.38**) and Poor (**6.53**) credit score holders.
- **Statistical Significance:** The p-value is < 0.0001, showing a significant difference in the number of bank accounts across credit score groups.
- **Implication:** It is possible that customers with Good credit scores manage fewer accounts more effectively, whereas those with Poor credit scores might hold more accounts, potentially leading to financial fragmentation. This suggests that simplicity in managing finances may contribute to a better credit score.

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9. Number of Credit Cards

- **Insight:** Individuals with a Good credit score hold fewer credit cards on average (**4.14**) than those with Standard (**5.40**) or Poor (**6.63**) credit scores.
- **Statistical Significance:** The p-value is < 0.0001, indicating that the differences in the number of credit cards held are statistically significant.
- **Implication:** Fewer credit cards could indicate a more disciplined approach to credit, which is linked to higher credit scores. Individuals with Poor credit scores may have multiple credit cards, which could result in higher debt and a lower credit score.

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10. Number of Credit Inquiries

- **Insight:** The average number of credit inquiries made by lenders on customers with a Poor credit score is significantly higher (**8.15**) compared to those with a Standard (**5.31**) and Good (**3.28**) credit score.
- **Statistical Significance:** The p-value is < 0.0001, suggesting that the number of credit inquiries is significantly related to the credit score.
- **Implication:** Frequent credit inquiries often occur when individuals are in financial distress or seeking credit due to low scores. This is reflected in the higher number of inquiries for those with Poor credit scores.

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11. Number of Delayed Payments

- **Insight:** The number of delayed payments is highest for customers with a Poor credit score (**15.85**), followed by those with a Standard score (**13.55**) and the fewest for those with a Good score (**8.57**).
- **Statistical Significance:** The p-value is < 0.0001, showing a significant relationship between delayed payments and credit score.
- **Implication:** Delayed payments significantly hurt credit scores. The Poor credit score category shows the highest number of delayed payments, which likely results in a more negative credit score.

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12. Outstanding Debt (USD)

- **Insight:** Individuals with a Poor credit score have the highest outstanding debt (**$2,081.47**) compared to those with a Standard (**$1,278.42**) and Good (**$801.26**) credit score.
- **Statistical Significance:** The p-value is < 0.0001, indicating a significant difference in outstanding debt across the credit score categories.
- **Implication:** A high level of outstanding debt can significantly lower a customer's credit score. Those with Poor credit scores have the highest debt, reinforcing the idea that high debt levels are a key factor in poor creditworthiness.

Conclusions & Recommendations:

1. **Financial Discipline:** Customers with a Good credit score generally show better financial management in terms of income, timely payments, and fewer delayed payments.
2. **Income and Debt:** A higher annual income and lower outstanding debt are strongly associated with better credit scores. Encouraging financial stability through higher income and controlled debt could help improve credit scores.
3. **Credit Behavior:** Managing fewer bank accounts, credit cards, and credit inquiries is linked to higher credit scores. Financial simplicity and fewer credit obligations may aid in maintaining a Good credit score.
4. **Risk Management:** Credit inquiries and delayed payments are warning signs of financial instability. Individuals with frequent credit inquiries and delayed payments may need targeted interventions to avoid further damage to their credit score.

Part 2

Objective: In this section, we explore the relationships between the **class variable Credit_Score** (Good, Standard, Poor) and other **categorical attributes** such as Occupation and various Loan Types using contingency tables. All relationships are statistically significant (based on p-values < 0.05).

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1. Credit_Score and Occupation:

	Good	Poor	Standard
Accountant	1205	2037	3502
Architect	1233	1855	3736
Developer	1210	1924	3586
Doctor	1164	1804	3600
Engineer	1264	2094	3506
Entrepreneur	1185	2033	3430
Journalist	1254	1840	3442
Lawyer	1259	1965	3872
Manager	1190	1856	3386
Mechanic	1107	2044	3625
Media_Manager	1244	1764	3712
Musician	1192	1786	3374
Scientist	1201	2062	3481
Teacher	1224	2027	3421
Writer	896	1907	3501

- Occupation appears to influence the distribution of credit scores (Good, Poor, and Standard).
- Some professions (e.g., Scientist, Teacher, Lawyer) have a relatively higher count in the Good credit score category, while others (e.g., Writer, Mechanic, and Entrepreneur) are more evenly spread between Poor and Standard.
- Engineer and Scientist categories have significant concentrations in the Poor category, indicating that these occupations tend to have lower credit scores.

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2. Credit_Score and Payment_of_Min_Amount:

- Credit_Score: Credit score assigned to the customer for given month
- Payment_of_Min_Amount: Whether minimum amount was paid by the customer or not

```
> contingency_tables_with_totals[["Payment_of_Min_Amount_Credit_Score"]]
$contingency_table

       Good  Poor Standard
  No  15620  5439    19509
  Yes  2208 23559    33665
```

- Most customers with **Good** credit scores did **not pay the minimum amount** (15,620 vs. 2,208 who did).
  - This might seem counterintuitive at first glance because good credit scores are typically associated with responsible financial behavior.

- Possible explanation: Customers with good scores might pay more than the minimum amount or even pay off their full balance, which still counts as "No" under the "Payment_of_Min_Amount" metric.

- The majority of customers with Poor credit scores paid the minimum amount (23,559 vs. 5,439 who did not).
  - This suggests that these customers are struggling financially but are making the minimum payments to avoid penalties or maintain some level of creditworthiness.
- Most customers with Standard credit scores also paid the minimum amount (33,665 vs. 19,509 who did not).
  - This aligns with expectations for this group, as they fall between good and poor credit, and their payment behavior reflects a mix of financial responsibility and challenges.

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3. Credit_Score and Credit_Mix:

- Credit_Score: Credit score assigned to the customer for given month
- Credit_Mix: Type of the mix of credits that the customer has

```
> contingency_tables_with_totals[["Credit_Mix_Credit_Score"]]
$contingency_table

            Good  Poor Standard
  Bad        351 14289     9128
  Good     14848  4851    10685
  Standard  2629  9858    33361
```

- For Good credit scores, the Good credit mix dominates, followed by Standard and Bad credit mixes.
- Poor credit scores show a dominance in Bad and Standard credit mixes, suggesting individuals in this group may have more difficulty managing their credit mix effectively.
- The Standard credit category has a substantial number of people with Standard credit mixes, and also with Bad credit mixes, showing a more mixed financial behavior in this group.

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4. Credit_Score and Last_Loan_1:

```
> contingency_tables_with_totals[["Credit_Score_Last_Loan_1"]]
$contingency_table

           Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
  Good          1442                1646                   1578             1537          1632    3444          1780        1643
  Poor          2883                3089                   3105             2957          3022    1336          2982        3142
  Standard      5003                5065                   5285             5306          4954    6628          5254        5319

           Personal Loan Student Loan
  Good              1472         1654
  Poor              3237         3245
  Standard          5227         5133
```

- Good credit scores show higher counts across multiple loan types, including Auto Loan, Credit-Builder Loan, and Mortgage Loan.
- For Standard credit scores, the distribution of loan types is spread across more loan categories, but No Loan and Payday Loan are the highest.
- Personal Loan and Student Loan appear with a higher frequency for individuals with Poor credit scores, suggesting that people with these scores may rely on loans like personal or student loans.

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5. Credit_Score and Last_Loan_2:

```
> contingency_tables_with_totals[["Credit_Score_Last_Loan_2"]]
$contingency_table

          Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
Good           1189               1170                   1079            1407          1225    6614          1371        1226
Poor           3032               2881                   2834            2905          2836    2569          2882        3263
Standard       4251               4477                   4359            4464          4347   13353          4475        4615

          Personal Loan Student Loan
Good               1235         1312
Poor               2891         2905
Standard           4514         4319
```

- Good credit scores again show a concentration in several loan categories, especially Auto Loan and Mortgage Loan.
- Poor credit scores are notably higher in the No Loan and Payday Loan categories.
- The Standard credit group has a wide distribution across loan types, with notable counts for Auto Loan, Credit-Builder Loan, and Debt Consolidation Loan.

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6. Credit_Score and Last_Loan_3:

```
> contingency_tables_with_totals[["Credit_Score_Last_Loan_3"]]
$contingency_table

          Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
Good            871                950                    807             828           814   10055           859         869
Poor           2354               2639                   2732            2371          2507    6326          2690        2577
Standard       3583               3731                   3389            3441          3551   21867          3531        3322

          Personal Loan Student Loan
Good               844          931
Poor              2331         2471
Standard          3233         3526
```

- The distribution for Good credit scores shows higher counts in the No Loan and Auto Loan categories, while Poor credit scores are more concentrated in No Loan and Debt Consolidation Loan categories.
- The Standard credit score category again shows a balanced spread, particularly with a higher concentration in Credit-Builder Loan and No Loan.

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7. Credit_Score and Last_Loan_4:

```
> contingency_tables_with_totals[["Credit_Score_Last_Loan_4"]]
$contingency_table

           Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
  Good           393                 565                     496              382           519   13657           412         508
  Poor          2057                2250                    2026             2264          2139    9763          2266        2124
  Standard      2270                2849                    2510             2554          2510   30580          2442        2656

           Personal Loan Student Loan
  Good               442          454
  Poor              1944         2165
  Standard          2470         2333
```

- The trend continues from previous loan categories, with Good credit scores more evenly distributed across loans like Auto Loan, Personal Loan, and Home Equity Loan.
- Poor credit scores continue to have higher frequencies in the No Loan and Personal Loan categories.
- Standard credit scores show a broader distribution of loans, with notable counts in Credit-Builder Loan and Debt Consolidation Loan, and a high frequency in No Loan.

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8. Credit_Score and Last_Loan_5:

- For Good credit scores, the distribution is similar, but the frequencies for certain loans, like No Loan, Personal Loan, and Credit-Builder Loan, are lower than in previous tables.
- Poor credit scores continue to show high frequencies in the No Loan and Auto Loan categories.
- Standard credit continue to score high in No Loan, scores also exhibit a spread in Auto Loan and Debt Consolidation Loan, but again show relatively lower counts in Personal Loan compared to earlier loan tables.

```
> contingency_tables_with_totals[["Credit_Score_Last_Loan_5"]]
$contingency_table

           Auto Loan Credit-Builder Loan Debt Consolidation Loan Home Equity Loan Mortgage Loan No Loan Not Specified Payday Loan
  Good            74                 90                     85              71            89   17130            91          76
  Poor          1908                1850                   1796            1822          1822   13047          1708        1755
  Standard      1442                1556                   1647            1507          1529   39279          1473        1513

           Personal Loan Student Loan
  Good              69           53
  Poor            1645         1645
  Standard        1806         1422
```